# ScanCrypt

Post-Quantum Cryptographic Security Assessment

| | |
|---|---|
| **Project Name** | m4fstack |
| **Analysis Date** | 2026-07-09 20:57:24 |
| **ScanCrypt Version** | 1.0 |
| **Algorithm** | ML-KEM / Kyber |
| **Security Level** | Unknown ML-KEM Variant |
| **Implementation Type** | Masked Implementation |

**Medium**

## Confidentiality Notice

This document contains confidential assessment details based on ScanCrypt analysis. Unauthorized disclosure is prohibited.

Report generated automatically by ScanCrypt.

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# Executive Summary

| | | | |
|---|---|---|---|
| **Overall Risk** | Medium | **Algorithm** | ML-KEM / Kyber |
| **Security Level** | Unknown ML-KEM Variant | **Implementation Type** | Masked Implementation |
| **Programming Language** | C/C++ | **Build System** | Unknown |
| **Files** | 3 | **Source Files** | 2 |
| **Header Files** | 1 | **Lines of Code** | 906 |
| **Rules Executed** | 0 | **Rules Passed** | 0 |
| **Rules Failed** | 0 | **Analysis Confidence** | 99% |
| **Detection Confidence** | 99% | **Analysis Duration** | N/A |



# Project Inspection

| | |
|---|---|
| **Algorithm** | ML-KEM / Kyber |
| **Security Level** | Unknown ML-KEM Variant |
| **Implementation Type** | Masked Implementation |
| **Programming Language** | C/C++ |
| **Build System** | Unknown |
| **Files** | 3 |
| **Source Files** | 2 |
| **Header Files** | 1 |
| **Lines of Code** | 906 |
| **Detection Confidence** | 99% |

## Detection Evidence

- poly detected in poly.h
- KYBER_N detected in poly.h
- Poly compression/decompression routines detected in poly.h
- NTT implementation detected in poly.h
- poly detected in poly.c
- NTT implementation detected in poly.c
- KYBER_K detected in poly.c
- KYBER_Q detected in poly.c
- KYBER_N detected in poly.c
- KYBER_ETA1 detected in poly.c
- KYBER_ETA2 detected in poly.c
- Poly compression/decompression routines detected in poly.c
- barrett_reduce detected in poly.c
- poly detected in indcpa.c
- NTT implementation detected in indcpa.c
- KYBER_K detected in indcpa.c
- IND-CPA primitives detected in indcpa.c
- Poly compression/decompression routines detected in indcpa.c
- Masked Implementation indicators detected in /tmp/scancrypt_upload_1ha2xf0s/poly.c
- KYBER_K was detected, but its value could not be extracted.

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- Build system not detected

# Findings Summary

| Rule ID | Severity | Category | Status | File | Line |
|---|---|---|---|---|---|
| No findings | | | | | |

# Detailed Findings

No findings were detected during analysis.

# Rule Execution Summary

No rule execution data available.

# Statistics

**Total rules executed**
0

**Passed rules**
0

**Failed rules**
0

**Analysis duration**
N/A

**Files analysed**
3

**Source lines analysed**
906

## Execution Progress

Passed: 0 / 1

Failed: 0 / 1

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# Appendix

## Analysed Files

m4fstack/poly.h

m4fstack/poly.c

m4fstack/indcpa.c

**Tool version**
1.0

**Report generated**
2026-07-09 20:57:24

**Scan configuration**
Default ScanCrypt profile

**Supported algorithms**
ML-DSA (Dilithium), ML-KEM (Kyber)

**Supported implementation types**
Portable C / Reference, AVX2 Optimized, ARM Cortex-M / Embedded, Masked Implementation, Hardware / FPGA

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